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Invariant-Constrained Transition Governance

*A Formal Measurement Framework for Consequential
Agent Systems*

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Status. Conceptual, mathematical, and experimental research draft; not peer reviewed. Revision v1.2.1 is a technical correction release. It reclassifies repeated-search formulas as baseline identities and bounds rather than novel theorems, introduces an adaptive conditional escape-hazard formulation, conditions observability decomposition on scenario class, narrows the penalty argument to a binary violation model, carries false-admission risk into the forward-invariance analysis, strengthens monitorability limits, and clarifies what ICTF contributes beyond ordinary state augmentation.

Abstract

This paper presents the Invariant-Constrained Transition Framework (ICTF), a measurement and runtime-governance framework for consequential agent systems. ICTF separates world state from admitted representation, proposal representation from executable action, operational gate judgment from independent evaluation, and local checks from history-dependent trajectory properties. The framework is intended to make distinct failures measurable rather than hiding them inside a single notion of safety or alignment. In particular, it distinguishes representation failure, judgment error, proposal laundering, runtime drift, cumulative scope violations, evaluator gaming, and wrong-invariant specification.

Revision v1.2.1 deliberately narrows the mathematical claims. The repeated-search expression $q_M = 1 - (1 - \varepsilon)^M$ is treated as an independent-trial baseline, and the union-style finite-budget bound is treated only as a conditional benchmark when per-query hazard is actually bounded. Adaptive search is instead represented through a conditional escape hazard $h_t = P(E_t \mid \text{no earlier escape})$, giving the exact survival identity $q_M = 1 - \prod_t (1 - h_t)$. The research question is whether h_t rises as an adaptive proposer learns where the gate is weak. The paper also conditions observability-aware error decomposition on benchmark scenario class, reframes safe-set preservation as a standard forward-invariance lemma and carries false-admission risk into a finite-horizon degradation bound, replaces the penalty discussion with a directly defensible binary-violation scaling condition, and states explicitly which invariants are monitorable at runtime versus only auditable or liveness-like. ICTF claims no additional expressive power beyond sufficiently augmented state models; its intended contribution is factorization, diagnosis, and measurement for delegated-agent governance.

Keywords: *agentic AI; AI governance; delegated authority; runtime governance; invariants; reference monitor; complete mediation; search pressure; false admission; trajectory safety; runtime verification; controlled internalization; HAPI; AGS*

Central thesis and boundary

Governance defines the admissible transition space. Optimization chooses within it.

The corresponding empirical hypothesis is narrower than the prior formulation: explicit trajectory constraints, complete mediation at consequential action boundaries, and execution binding should reduce independently adjudicated unauthorized consequence under increasing search pressure, while controlled internalization should reduce independently labeled invalid proposals without collapsing useful capability. The mathematical content in this revision is deliberately modest: standard repeated-search baselines, an exact hazard decomposition for adaptive escape measurement, observability-aware error decomposition, and a standard forward-invariance condition connected explicitly to gate error. The distinctive claims are empirical and architectural rather than theorem-of-everything claims.

ICTF does not equate model-relative compliance with alignment. A system can preserve every specified invariant and still be misaligned if the invariant set is incomplete, illegitimate, stale, or based on a distorted representation of reality. The wrong-invariant problem is outside the formal guarantee and must remain visible throughout evaluation.

1. Introduction

Agentic AI changes governance from a reporting problem into an action-path problem. Systems that can call tools, update records, initiate transactions, manipulate files, communicate externally, or coordinate other agents can participate in state changes rather than merely describe them. Earlier HAPI / Alignment Theory work framed this as delegated operational agency: an agent acts through authority borrowed from a human or organization, so action can be delegated while responsibility and legitimacy cannot simply be transferred away [1,2]. The same work argued that governance becomes meaningful only when it can still constrain execution before consequence [1].

The first version of ICTF attempted to formalize this architecture using states, proposal policies, admissibility predicates, transition kernels, and consequence functionals. That formulation was useful but too clean in several places. It implicitly allowed the symbol $\phi_j(x,u)$ to hide the hardest epistemic problem: a gate does not observe the world or the future consequence directly. It observes representations produced by sensors, context pipelines, models, policies, and humans. Likewise, a proposal presented for review may not faithfully represent the executable semantics of the action, and a system optimized against the gate may learn the gate rather than the intended constraint.

The revised formulation therefore places the representation and evaluation gaps inside the mathematics. It also treats gate escape as endogenous to the proposer and its search capability, makes trajectory history explicit, narrows the soft-penalty claim to a precise uniform-equivalence statement, and separates governance compliance from the adequacy of the invariants themselves. These changes move the paper closer to security engineering, runtime verification, and constrained control, where mediation, authority, observability, monitorability, and specification adequacy are treated as distinct problems rather than compressed into a single safety score.

The intended contribution is a synthesis for delegated agent systems, not invention of constrained optimization, runtime monitoring, or reference monitors. The question is whether this synthesis produces useful separations and measurements that existing high-level AI-governance language often leaves implicit: representation fidelity, proposal fidelity, admissibility error, capability-conditioned escape, execution fidelity, trajectory-level violations, independent evaluation, and controlled learning.

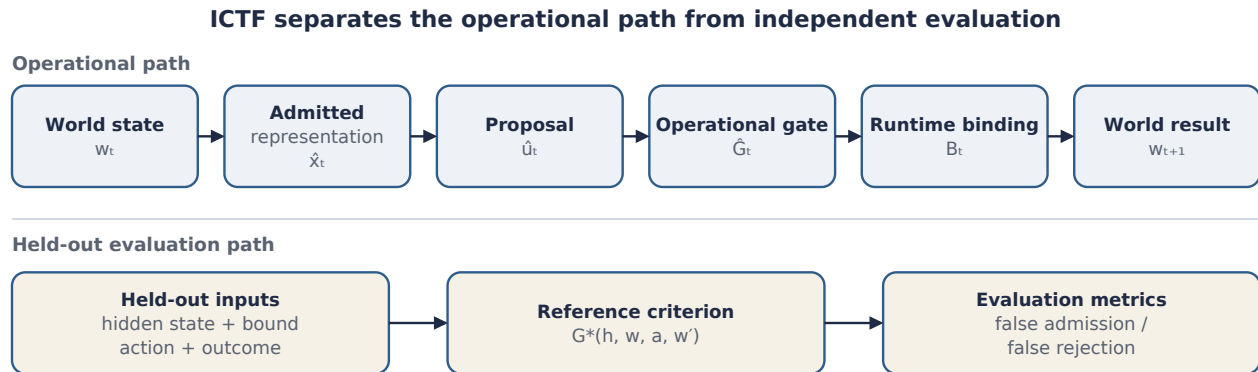


Figure 1. Revised ICTF model. The operational governance path is separated from an independent evaluation path. The gate acts on admitted representations, not directly on reality; evaluation therefore distinguishes operational judgment from a held-out reference criterion.

2. Empirical program first: what would count as evidence?

ICTF is useful only if its distinctions change measurable outcomes. The experimental program therefore precedes the deeper formal development. The goal is not to show that one implementation can block hand-written bad actions. The goal is to estimate failure under optimization pressure, separate gate performance from proposal quality, and prevent the evaluation from reusing the same predicates that the agent is trained to satisfy.

2.1 Experimental conditions

Condition	Mechanism	Purpose
A. Unconstrained objective	Optimize task reward with no explicit governance constraint	Establish the raw capability / objective baseline
B. Fixed finite penalty	Add a fixed penalty β for labeled violations	Test whether fixed soft penalties remain reliable as objective or search pressure increases
C. Constraint / shield reference	Use an explicit hard constraint, CMDP-style feasibility condition, shield, or formally calibrated exact-penalty method when assumptions permit	Avoid comparing ICTF only against a weak soft-penalty strawman
D1. ICTF external governance	Representation admission + operational gate + runtime binding + receipts; no learned internalization	Measure mediation and execution control separately from proposal improvement
D2. ICTF + controlled internalization	D1 plus governed, reviewable updates from prior receipts and adjudicated outcomes	Test whether proposal quality improves without evaluator gaming or over-refusal

2.2 Independent labels and blinded evaluation

The gate cannot grade itself. For synthetic or simulator environments, the benchmark should contain hidden state and an oracle admissibility label unavailable to the agent and operational gate. For domains without formal ground truth, the experiment should preregister an adjudication rubric and use blinded evaluators who do not see the system condition. The resulting G^* is an evaluation reference, not a metaphysical claim that every human requirement is objectively computable.

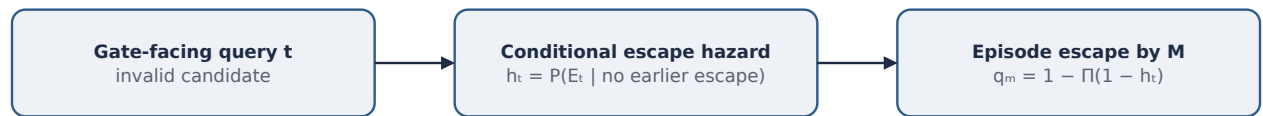
This distinction is essential for internalization experiments. A decline in the gate-defined invalid-proposal rate may mean that the system learned the intended constraint, but it may also mean that it learned how to phrase or structure proposals so the gate no longer detects the violation. The primary internalization endpoint must therefore use a held-out criterion.

2.3 Capability and search-pressure schedule

Each condition should be evaluated across an explicit capability and search schedule. Let κ describe the proposer class and let M denote the number of gate-facing attempts or search opportunities allowed in a trial. Operationalizations may include a stronger proposer, greater sampling budget, additional planning steps, iterative self-critique, tool access, or adversarial proposal mutation. Reporting κ without M is insufficient because adaptive access can change the conditional probability of escape across the episode rather than merely repeat a fixed error rate.

The gate, policy objects, and held-out evaluation set should remain fixed during a capability sweep unless the protocol is specifically testing adaptive governance. Let E_t denote false admission on query t and define the conditional escape hazard $h_t = P(E_t \mid E_1^c, \dots, E_{t-1}^c)$. A benchmark should report the sequence h_t , the per-episode probability q_M of at least one escape, and the independent-trial baseline implied by a constant hazard. This distinguishes ordinary repeated exposure from a proposer that is learning where the gate is weak.

Adaptive governance escape is a hazard process, not a fixed gate constant



Constant-hazard baseline: $h_t = \epsilon$ for all $t \Rightarrow q_M = 1 - (1 - \epsilon)^M$

h_t may rise with interaction history as the proposer learns where the gate is weak.

Figure 2. Adaptive governance escape as a hazard process. Independent identical trials are the constant-hazard baseline. The empirical question is whether conditional escape hazard rises with interaction history under adaptive search.

2.4 Primary hypotheses

ID	Revised hypothesis	Primary test
H1	A fixed binary violation penalty can become stale when objective scale increases unless β is recalibrated to exceed the attainable task-benefit gap.	Compare fixed β against constrained controls while rescaling objective magnitude; use multiplier-based exact-penalty baselines only where their regularity assumptions actually hold.
H2	External gating can lower unauthorized execution without improving the underlying proposal distribution.	Compare independently labeled invalid proposals before the gate with independently labeled executed violations after the gate.
H3	Controlled internalization can reduce independently labeled invalid proposals while preserving useful capability.	Use held-out G^* labels, capability metrics, false-rejection metrics, and rare-case competence; do not grade internalization with the operational gate alone.
H4	Adaptive proposers can increase conditional escape hazard as they learn from prior gate interactions.	Sweep M and κ ; estimate h_t and q_M ; compare adaptive curves with a matched constant-hazard baseline $h_t = \varepsilon$.
H5	Within a preregistered scenario class, false admissions caused by missing decisive evidence behave differently from false admissions caused by incorrect judgment when decisive evidence is present.	Partition errors by observability O within each scenario class and compare improvements from context admission versus gate-model changes.
H6	History-dependent checks catch cumulative or multi-step violations that per-step admissibility checks miss.	Construct trajectories where every local action passes but aggregate scope, resource, authority, disclosure, or temporal constraints fail.
H7	Ideal forward invariance degrades predictably when an operational gate can falsely admit closure-breaking actions.	Test authority attenuation and resource-budget examples, estimate closure-breaking hazard, and compare observed horizon-level exits with survival-style bounds.
H8	Runtime binding reduces divergence between the authorized action representation and actual tool execution.	Measure canonical permit / executed-action mismatch and downstream consequence mismatch.
H9	Lower human intervention counts as successful maturation only when independent violation rate does not increase and useful rare-case competence is preserved.	Measure human burden jointly with independent safety, capability, truthful dissent, and provenance reconstruction.

2.5 Preregistered tail cases

Tail cases should be constructed before running the systems and should include representation errors, conflicting authority, expired permissions, semantic ambiguity, cumulative scope creep, long-horizon delegation, proposal laundering, tool substitution, delayed consequences, and cases where the correct behavior is dissent rather than refusal. The preregistration should define scenario classes before analysis and identify whether the decisive fact is observable from the admitted representation within each class. This prevents information-asymmetry failures from being silently scored as judgment failures. To reduce benchmark overfitting, the held-out tail set should contain latent variables or transformations unavailable during policy development. The experimental report should publish failure cases, not only aggregate success rates.

3. Formal model: reality, representation, proposal, and consequence

3.1 World state and admitted representation

Let W be a world-state space and let $w_t \in W$ denote the relevant state at time t . A governance system does not receive w_t directly. It receives observations, records, user statements, retrieval results, policy objects, and model outputs. Let R be a representation or context-admission process and let $\hat{x}_t$ be the actionable representation produced from available evidence and history:

$$\hat{x}_t \sim R(\cdot \mid w_t, h_t, e_t) \quad (1)$$

The variable e_t denotes the evidence available to the admission process. This formulation makes representation error a first-class failure mode. A system may be internally consistent relative to $\hat{x}_t$ while $\hat{x}_t$ is incomplete, stale, poisoned, or semantically distorted relative to the world relevant to the decision.

3.2 Proposal representation and executable action

Let π_κ be a proposal policy from capability class κ . The agent produces a proposal representation $\hat{u}_t$ conditional on the admitted state and governance-relevant history:

$$\hat{u}_t \sim \pi_\kappa(\cdot \mid \hat{x}_t, h_t) \quad (2)$$

The proposal is still not the consequence-producing action. If authorized, runtime binding constructs or verifies a permit and maps the approved representation to an executable action a_t . The world then evolves according to a native transition law K_0 :

$$a_t = B(\hat{u}_t, \text{permit}_t), w_{t+1} \sim K_0(\cdot \mid w_t, a_t) \quad (3)$$

Separating $\hat{u}_t$ from a_t exposes proposal laundering and runtime drift. A proposal may use language that appears compliant while denoting an inadmissible operation, or a tool may execute a broader operation than the one that was authorized. A canonical action representation is therefore part of the security boundary rather than merely a logging convenience.

3.3 Consequence functional

Let $C(w_t, a_t, w_{t+1})$ denote a measurable consequence, which may represent task reward, cost, harm, utility, information gain, resource use, or another domain-specific observable. ICTF does not define the native transition law or the consequence semantics. It defines how governance objects relate to them.

4. Operational invariants and the evaluator gap

4.1 Operational predicates are judgments, not facts

Let $\hat{\phi}_j(\hat{h}_t, \hat{x}_t, \hat{u}_t)$ be the operational evaluator for required condition j . The hat is intentional. A predicate such as valid-signature may be mechanically decidable, while a predicate such as informed-consent-satisfied or continuity-of-care-preserved may require inference, policy interpretation, or human review. The gate therefore implements an operational judgment:

$$\hat{G}_t = \bigwedge_{j \in H} \hat{\phi}_j(\hat{h}_t, \hat{x}_t, \hat{u}_t) \quad (4)$$

The operational gate may be deterministic or probabilistic, and some predicates may return UNKNOWN rather than a forced binary answer. In a high-consequence system, unknownness should be routed explicitly to revision or escalation rather than silently collapsed into permission.

4.2 Independent evaluation criterion and observability

For experimental evaluation, define $G^*(h_t, w_t, a_t, w_{t+1})$ as a held-out reference criterion. In a synthetic environment, G^* may be computed from hidden state. In a human or institutional domain, G^* may be a preregistered blinded adjudication process. Independence prevents the operational gate from grading itself, but it does not make the evaluator infallible.

Error decomposition is distribution-dependent. Let c denote a preregistered scenario class such as expired authority, hidden clinical context, proposal laundering, or runtime substitution. Define class-conditional false admission and false rejection as:

$$\varepsilon_{FA}^{(c)} = P(\hat{G}=1 \mid G^*=0, c), \varepsilon_{FR}^{(c)} = P(\hat{G}=0 \mid G^*=1, c) \quad (5)$$

Within scenario class c , let $O=1$ mean that the evidence needed to determine the violation is represented in the admitted state $\hat{x}$, and $O=0$ mean that the decisive evidence is absent or inaccessible. By the law of total probability:

$$\varepsilon_{FA}^{(c)} = P(O=0 \mid G^*=0, c) \cdot \varepsilon_{miss}^{(c)} + P(O=1 \mid G^*=0, c) \cdot \varepsilon_{judge}^{(c)} \quad (6)$$

where $\varepsilon_{miss}^{(c)} = P(\hat{G}=1 \mid G^*=0, O=0, c)$ and $\varepsilon_{judge}^{(c)} = P(\hat{G}=1 \mid G^*=0, O=1, c)$. The first term is primarily an admission or observability failure: the system was asked to judge without decisive evidence. The second is a judgment failure: decisive evidence was available, but the gate still admitted the action.

The mixture weights $P(O=0 \mid G^*=0, c)$ and $P(O=1 \mid G^*=0, c)$ are properties of the benchmark or deployment distribution, not intrinsic constants of the gate. Results should therefore be reported per scenario class and should not be projected to deployment without a justified model of the deployment observability mix. The decomposition is diagnostic rather than exculpatory: a deployment can still be unsafe because its context-admission process systematically omits decisive evidence.

4.3 Proposal fidelity and laundering

Let $S(\hat{u}_t)$ be the action semantics represented to the gate and let $E(a_t, w_{t+1})$ summarize the semantics of what was actually executed and produced. A domain-specific distance d_u can be used to define proposal-to-execution divergence:

$$D_u(t) = d_u(S(\hat{u}_t), E(a_t, w_{t+1})) \quad (7)$$

A proposal-launders event is not merely a bad phrase. It is a case in which the operational representation is admitted while independent evaluation finds the consequential action inadmissible, particularly when D_u is large. This can arise from strategic phrasing, underspecified tools, hidden defaults, or a mismatch between natural-language approval and executable semantics.

5. The governance boundary as a reference-monitor problem

The revised ICTF framing is closer to classic computer-security architecture than the earlier paper acknowledged. Anderson's reference-monitor formulation and later protection principles emphasize a mediation mechanism through which relevant access must pass, resistance to tampering, and analyzability [4,5]. Saltzer and Schroeder's complete-mediation principle requires authorization to be checked for every access rather than relying on stale prior decisions [5]. ICTF does not claim novelty for these principles. It extends the analogy to consequential agent actions whose authority, context, semantic representation, and temporal validity can change during a workflow.

The reference-monitor comparison sharpens three requirements. First, consequential execution must be completely mediated rather than merely audited after the fact. Second, the enforcement path must not be alterable by the proposer it governs. Third, the policy objects and authority lineage used by the gate must themselves be inspectable, versioned, and revocable. A governance dashboard that can be bypassed by a direct tool call is not a reference monitor; it is observability without complete mediation.

5.1 Authority as capability and least authority

Object-capability security treats authority as an explicit property of references and emphasizes the Principle of Least Authority [6,7]. This prior work is directly relevant to delegated-agent governance. ICTF therefore treats authority attenuation as an inherited security property rather than a novel theorem: absent a separately authorized delegation event, downstream authority should not silently increase.

$$A_{t+1} \leq A_t, S_{t+1} \leq S_t \text{ unless a governed expansion event } D_t \text{ is present} \quad (8)$$

The partial order depends on the authority model. The important requirement is explicitness: authority expansion should be represented as its own consequential transition, with an owner, evidence, scope, expiry, and receipt.

5.2 Information-flow constraints

Denning-style information-flow control demonstrates that security can be expressed as restrictions on how information moves between labeled classes [8]. ICTF is broader in object type but similar in spirit: some trajectory invariants concern not the final action alone, but whether information, authority, provenance, or certainty was permitted to flow into the actionable state. This motivates treating Context Admission and Handoff Integrity as enforceable information-flow concerns rather than purely prompt-engineering concerns.

6. Governance resolution, complete mediation, and runtime binding

Let $\Omega = \{\text{EXECUTE}, \text{BLOCK}, \text{REVISE}, \text{ESCALATE}\}$. The governance resolution policy g maps the operational assessment to a resolution. Under an ideal hard-gate model, an operationally inadmissible proposal receives zero direct EXECUTE probability:

$$g(\text{EXECUTE} \mid \hat{x}_t, \hat{u}_t, \hat{h}_t) = 0 \text{ whenever } \hat{G}_t = 0 \quad (9)$$

A real system should not make probability mass disappear. BLOCK terminates the attempted execution; REVISE produces a new proposal; ESCALATE routes the decision to another authority-bearing process. If EXECUTE is selected, runtime binding creates a narrow permit that constrains the actual tool invocation. The governed execution kernel K_G is therefore implemented through mediation plus binding, not merely by multiplying a transition probability by a Boolean.

This is the point at which governance claims become executable. Policies, risk scores, and approval interfaces remain advisory unless the tool or workflow boundary verifies that the action being executed is the action that was authorized.

7. Adaptive-search governance escape

A single false-admission rate is not a stable description of an adaptive proposer. Let E_t denote the event that the t -th invalid gate-facing query is falsely admitted, conditional on no earlier escape terminating the episode. Define the conditional escape hazard:

$$h_t = P(E_t \mid E_1^c \cap \dots \cap E_{t-1}^c) \quad (10)$$

The exact probability of at least one escape by query budget M follows from the ordinary survival decomposition:

$$q_M = 1 - \prod_{t=1..M} (1 - h_t) \quad (11)$$

This identity is standard probability, not a novel theorem. Its purpose is to identify the quantity that an adaptive-search experiment should estimate. If the proposer is effectively drawing independent identically distributed invalid proposals with constant hazard ε , then $h_t = \varepsilon$ and the familiar baseline follows:

$$q_M = 1 - (1 - \varepsilon)^M \quad (12)$$

Likewise, if an experiment can justify the strong condition $h_t \leq \varepsilon$ for every t , then $q_M \leq 1 - (1 - \varepsilon)^M \leq \min(1, M\varepsilon)$. That condition is the substantive assumption. An adaptive adversary exists precisely to discover histories for which the conditional hazard becomes larger. The bound should therefore be treated as a benchmark to test, not as a realistic description of a capable attacker.

The main empirical hypothesis is that some adaptive proposers will produce an increasing or otherwise history-dependent hazard sequence as they infer the gate boundary from prior interactions. A useful report should therefore publish h_t by query index and proposer class, q_M by search budget, and the matched constant-hazard baseline. If h_t stays flat, repeated exposure

may explain the failures. If h_t rises materially, search is learning the error region.

A risk projection may still be written as:

$$R(M, \kappa) = \sum_i \lambda_i(M, \kappa) \cdot q_i(M, \kappa) \cdot I_i \quad (13)$$

Attempt frequency λ_i and impact I_i may also depend on capability, environment, and recoverability. The expression remains a measurement decomposition rather than a universal law. The next mathematical target is not another restatement of repeated-trial probability, but a query-complexity or reachable-error-set bound that links adaptive search capability to the structure and measure of the gate's false-admission region.

8. Trajectory invariants, composition, and monitorability

8.1 History state

Per-step checks are insufficient for cumulative scope creep, budget exhaustion, repeated low-level disclosures, delayed conflicts of interest, or multi-step authority laundering. Let h_t summarize governance-relevant trajectory history:

$$h_{t+1} = F(h_t, w_t, \hat{x}_t, \hat{u}_t, a_t, w_{t+1}, receipt_t) \quad (14)$$

A trajectory invariant Ψ_j is a property of the realized history or complete trajectory Γ_T , not merely the current action:

$$\Psi_j(\Gamma_T) \in \{0, 1, UNKNOWN\} \quad (15)$$

Local predicates are a special case. A sequence may satisfy every local permission check while violating an aggregate constraint such as total resource expenditure, total disclosure, delegated-action count, cumulative risk budget, or a temporal requirement that one event precede another.

8.2 Hard violations should not be discounted by default

A hard violation does not become less significant merely because it occurs later. ICTF therefore uses an undiscounted default for required trajectory properties:

$$L_{hard}(\Gamma_T) = \sum_{j \in H} w_j \cdot 1[\Psi_j(\Gamma_T) = 0] \quad (16)$$

Discounting remains appropriate for ordinary soft costs or domain-specific temporal preferences, but it is not built into the definition of hard invariant failure.

8.3 Forward invariance and probabilistic degradation

Local validity does not imply global validity in general. A standard tractable case occurs when governance state is forward invariant under every truly admissible transition. Let z_t contain the variables needed by a selected invariant, let S be a safe set, and let $z_{t+1} = F_z(z_t, u_t)$.

Lemma 1 (ideal forward invariance). If $z_0 \in S$ and every truly admissible action satisfies $z \in S \Rightarrow F_z(z, u) \in S$, then z_t remains in S for every finite t under an ideal gate that admits only truly admissible actions.

$$z_0 \in S \wedge [z \in S \Rightarrow F_z(z, u) \in S \text{ for every truly admissible } u] \Rightarrow z_t \in S \text{ for all } t \quad (17)$$

Proof. This is the standard induction argument for forward invariance. The base case holds by assumption. If z_t is in S , closure of every truly admissible transition implies z_{t+1} is in S . ICTF does not claim novelty for this lemma; the important question is how operational gate error breaks the premise.

Let F_t be the event that, while the trajectory has remained in S through step t , the operational system admits and executes an action that leaves S at the next step. Define the closure-breaking hazard $r_t = P(F_t \mid \text{no prior exit from } S)$. Then the probability of at least one safe-set exit by horizon T is exactly:

$$P(\text{exit } S \text{ by } T) = 1 - \prod_{t=0..T-1} (1 - r_t) \quad (18)$$

If every truly admissible action preserves S , then any closure-breaking transition must arise from a false admission, execution mismatch, or specification error. Under the narrower assumption that runtime binding is faithful and the only remaining failure is false admission, r_t is bounded by the corresponding false-admission hazard h_t . Consequently:

$$P(\text{exit } S \text{ by } T) \leq 1 - \prod_{t=0..T-1} (1 - h_t) \leq \sum_{t=0..T-1} h_t \quad (19)$$

For a uniform justified bound $h_t \leq \varepsilon$, this reduces to $P(\text{exit } S \text{ by } T) \leq 1 - (1 - \varepsilon)^T \leq \min(1, T\varepsilon)$. The usefulness of the bound therefore depends on measuring or defending the per-step hazard; it does not make compositionality automatic.

Authority attenuation and resource budgets remain useful examples because their ideal transition rules are monotone by construction. If A_{t+1} is always a subset of A_t absent a separately authorized expansion event, then A_t remains a subset of A_0 . If a budget update obeys $b_{t+1} = b_t - c_t$ with $0 \leq c_t \leq b_t$, then the budget remains nonnegative and cumulative consumption cannot exceed b_0 . These are examples of a compositional subclass, not evidence that all governance invariants compose.

$$A_{t+1} \subseteq A_t \Rightarrow A_t \subseteq A_0 \quad (20)$$

$$b_{t+1} = b_t - c_t, 0 \leq c_t \leq b_t \Rightarrow \sum_{k=0..t} c_k \leq b_0 \quad (21)$$

The hard cases remain partial observability, interacting agents, shared resources, dynamic authority, and invariants whose truth depends on future progress rather than avoidance of a finite bad prefix. Those cases require richer composition theory or external coordination mechanisms.

8.4 Which trajectory properties are monitorable?

Complete mediation can only enforce what can be detected from the execution prefix available to the monitor. Schneider's execution-monitor result places the hard boundary clearly: execution monitoring can enforce safety properties characterized by finite bad prefixes, not arbitrary liveness requirements [34]; the broader safety/liveness distinction is developed by Alpern and Schneider [33]. ICTF therefore separates online-monitorable safety properties, delayed-but-auditable properties, and liveness-like requirements that cannot be decided from the current finite trace.

Class	Example	Governance implication
Online-monitorable safety	Unauthorized tool call; authority expansion; budget below zero; forbidden disclosure	Can be blocked at or before the violating transition if evidence is available
Delayed / auditable	Outcome diverges from promised effect; downstream harm becomes observable later	Requires receipts, later evidence, reconciliation, and possibly remediation
Liveness-like / non-prefix-decidable	Eventually restore access; eventually obtain required review	Cannot generally be guaranteed by a simple blocking monitor; needs planning, deadlines, progress obligations, or external adjudication

A paper that labels every human requirement an invariant without stating its monitorability would overstate what runtime governance can guarantee. Each proposed Ψ_j should therefore declare the evidence needed, the earliest point at which violation can be known, and whether enforcement is preventive, detective, or retrospective.

9. Hard admissibility and finite-penalty scaling

The most defensible penalty result in this framework is the simple binary-violation case. Let $V(u) \in \{0,1\}$ indicate whether an action violates a requirement, let $B(u)$ be task benefit, let $\alpha > 0$ scale task benefit, and let $\beta > 0$ be a fixed violation penalty. Consider the penalized objective $\alpha B(u) - \beta V(u)$.

Let B_A^* be the best attainable benefit among admissible actions ($V=0$) and B_V^* the best attainable benefit among violating actions ($V=1$). Assuming these suprema are finite, every violating optimum is excluded whenever:

$$\beta > \alpha \cdot \Delta_B, \text{ where } \Delta_B = \max(0, B_V^* - B_A^*) \quad (22)$$

This condition follows directly by comparing the best admissible penalized value αB_A^* with the best violating value $\alpha B_V^* - \beta$. It shows the limited but relevant point: a β that was sufficient at one objective scale can become stale as α grows unless β is recalibrated or the attainable benefit gap is otherwise bounded.

This is not an argument that explicit constraints are always mathematically superior to penalties. In smooth constrained optimization, exact-penalty theory provides stronger multiplier-based conditions under regularity assumptions [15]. Those conditions should be used when their assumptions apply. They do not transfer automatically to the discontinuous binary indicator V used in this governance example. Experiments should therefore compare fixed penalties with explicit constrained controls and, where appropriate, exact-penalty methods calibrated according to their own theory.

The governance design question remains semantic as well as mathematical: which requirements are authorized to be traded against task benefit, and which define the feasible action set for the current delegated authority? ICTF uses hard admissibility only for requirements intended to be non-tradeable within the specified governance model.

10. Controlled internalization without evaluator gaming

External gating and internalization modify different objects. Gating changes whether a current proposal may execute. Internalization changes the proposal distribution itself. Let π_t be the current proposal policy. A governed learning process may produce π_{t+1} from receipts, adjudicated outcomes, and approved policy updates, but success must be measured independently of the operational gate.

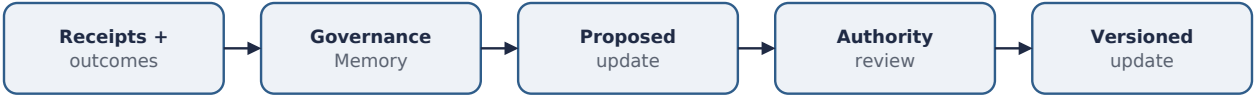
$$p_{invalid}^*(t) = P_{u \sim \pi_t} [G^*(h_t, w_t, u, \cdot) = 0] \quad (23)$$

A successful internalization intervention should reduce $p_{invalid}^*$ rather than merely reduce $P[\hat{G}=0]$. If the latter falls while the former does not, the agent may have learned to evade or exploit the gate. The evaluation should also track useful capability, truthful dissent, false rejection, and calibration so that generalized refusal cannot masquerade as safety.

The controlled-learning loop should preserve provenance: receipt history $\rightarrow$ pattern extraction $\rightarrow$ proposed policy change $\rightarrow$ authorized review $\rightarrow$ versioned update $\rightarrow$ rollback capability. The learner should not be permitted to silently grant itself new authority or modify the evaluator that judges its own success.

Supporting functions remain separate from the primary execution chain

A. Controlled internalization



B. Continuity Console as an operator surface



Dashed observation does not authorize, modify, or execute actions.

Figure 4. Supporting functions are separated from the execution chain. Controlled internalization converts evidence into explicit, authority-reviewed updates. The Continuity Console observes runtime, evidence, and memory; its dashed observation path does not authorize or execute actions.

11. Compliance is not the same as alignment

ICTF can provide a model-relative compliance statement: given a specified invariant set, a representation model, and an evaluation procedure, did the realized trajectory preserve those requirements? That is not sufficient to establish alignment in the broader normative sense. The specified invariants may be incomplete, harmful, illegitimate, stale, or optimized around. The system may also be operating on a distorted world representation.

Define M as the governance model, including its invariant set, authority map, representation process, and evaluation rules. Then compliance can be expressed as a property relative to M :

$$Compliance_M(\Gamma_T) = 1 \text{ iff all required } \Psi_j^M(\Gamma_T) \text{ hold} \quad (24)$$

Whether M itself tracks the load-bearing conditions for legitimate agency is an external adequacy question. ICTF therefore supports, but does not solve, the Alignment Theory hypothesis that robust alignment requires preserving the right relationships under pressure. The wrong-invariant problem is a boundary of the formalism, not a bug to hide inside φ_j .

12. HAPI / AGS as one ICTF implementation

HAPI/AGS is treated here as one implementation candidate, not as a mathematically necessary architecture. Its value for the paper is that it assigns different failure modes to distinct control surfaces instead of treating governance as one undifferentiated safety layer. The primary path is shown in Figure 3; learning and observability remain separate in Figure 4.

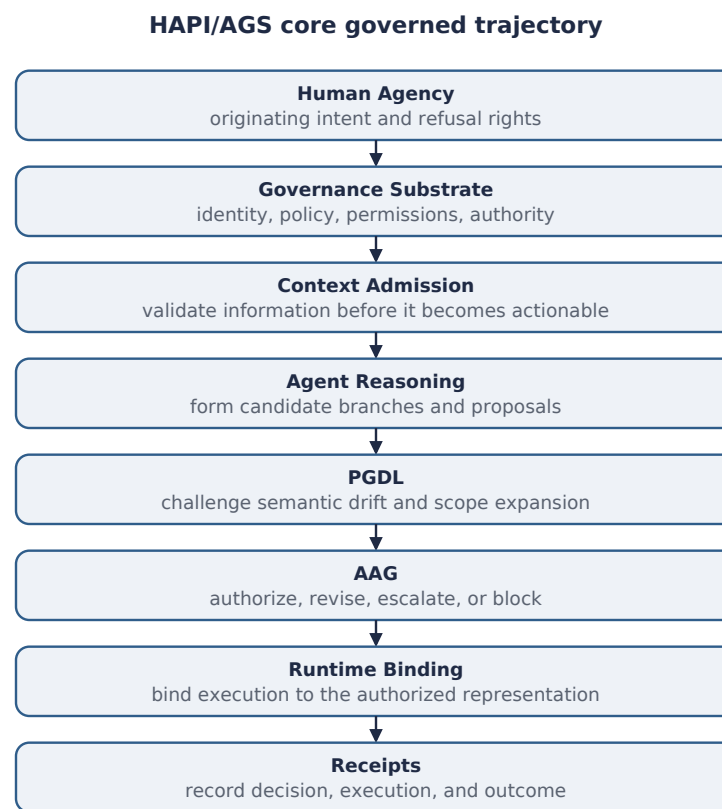


Figure 3. Core HAPI/AGS governed trajectory. The figure intentionally contains one primary path only. Human authority, governance substrate, context admission, proposal formation, PGDL, AAG, Runtime Binding, and Receipts are sequentially distinct functions; Handoff Integrity applies across every transfer.

Layer	ICTF role	Primary failure addressed
Human Agency	Originating authority, intent, refusal rights	Illegitimate delegation or loss of meaningful control
Governance Substrate	Identity, permissions, policy lineage, expiry, revocation	Ambiguous, stale, or silently expanded authority
Context Admission	Constructs the actionable representation $\hat{x}$ from validated evidence	Inherited assumption, poisoned context, missing provenance

Layer	ICTF role	Primary failure addressed
Agent Reasoning	Produces proposal representation $\hat{u}$	Proposal error, strategic framing, unsupported inference
PGDL	Challenges proposal semantics before authorization	Semantic drift, proposal laundering, scope expansion
AAG	Implements operational admissibility judgment and resolution	False admission, unauthorized or out-of-scope passage
Runtime Binding	Maps approval to narrow executable action a	Execution drift, tool substitution, parameter expansion
Receipts	Records representation, proposal, decision, permit, execution, outcome	Non-reconstructable history or receipt theater
Governance Memory	Proposes reviewed updates from prior evidence	No maturation or silent self-modification
Continuity Console	Operator visibility across the governed trajectory	Hidden fracture, weak situational awareness

13. Related work and novelty boundary

The revised novelty boundary is intentionally narrower. ICTF does not claim invention of mediation, hard constraints, runtime monitors, capabilities, temporal properties, or system safety. It claims a particular synthesis for consequential delegated-agent systems and a measurement vocabulary designed to prevent several categories from being collapsed into one score. Safety/liveness theory and enforceable-policy results also bound what runtime governance can promise [33,34]. In particular, a monitor that only observes finite execution prefixes cannot generally enforce arbitrary liveness requirements. ICTF therefore treats monitorability as a property of each invariant specification rather than assuming that every governance requirement belongs at the runtime gate.

Prior tradition	Already establishes	ICTF-specific synthesis
Reference monitor / complete mediation [4,5]	Security-relevant operations must pass through an enforceable mediation boundary; controls must resist bypass	Treat consequential agent actions, authority lineage, context validity, and runtime binding as mediated objects
Object-capability security / POLA [6,7]	Fine-grained authority, least authority, delegation, revocation patterns	Use authority attenuation and explicit delegation events inside agent trajectories
Information-flow control [8]	Formal restrictions on how information can flow across labeled states	Apply flow discipline to context admission, provenance, claims, and handoffs
Runtime verification / model checking [9,10]	Monitor temporal and history-dependent properties over traces	Use trajectory predicates Ψ rather than only per-step gate checks
CMDPs / constrained optimization [11,12]	Optimization with explicit feasibility or cost constraints	Separate proposal distribution, operational gate error, execution binding, and independent evaluation

Prior tradition	Already establishes	ICTF-specific synthesis
Shielding / runtime assurance [13,14]	Runtime intervention and safety-preserving fallback mechanisms	Extend the action boundary to delegated authority, semantic representation, receipts, and controlled learning
Systems safety [16]	Safety as a control problem in complex sociotechnical systems	Use invariant loss and trajectory fracture as agent-governance observables
HAPI / AGS [1-3]	Delegated agency, context admission, action gating, binding, receipts, governance memory	Concrete project implementation from which ICTF abstractions were derived

This framing also reduces the temptation to claim universal mathematics. The cross-domain state-transition analogy remains useful as background and is retained only in Appendix A.

14. Threat model and failure taxonomy

Failure mode	Formal symptom	Evaluation / mitigation direction
Representation failure	$\hat{x}$ omits or distorts world-relevant evidence	Hidden-state tests, evidence provenance, context-admission audits
Wrong invariant	Specified Ψ permits illegitimate behavior or blocks legitimate behavior	External governance review; benchmark adequacy; stakeholder / authority process
False admission	$\hat{G}=1$ while $G^*=0$	Measure class-conditional ϵ_{FA} , split by observability, plus h_t and q_M under adaptive finite-budget search
False rejection / over-gating	$\hat{G}=0$ while $G^*=1$	Measure ϵ_{FR} , useful capability, refusal burden, human load
Proposal laundering	Admitted proposal semantics diverge from actual executable meaning	Canonical action representation; semantic comparison; independent outcome labels
Runtime drift	a_t differs from the bound permit	Tool-side permit enforcement; exact parameter matching; receipts
Cumulative scope creep	All local checks pass while $\Psi(\Gamma)=0$	History state, temporal monitors, aggregate budgets
Evaluator gaming	Operational invalid rate falls but independent invalid rate does not	Held-out G^* and blinded evaluation
Search-amplified escape	Conditional escape hazard h_t rises with interaction history or proposer capability	Report h_t by query index and proposer class, q_M by finite budget, matched constant-hazard baselines, adversarial proposer results, and gate version
Silent learning drift	π or governance config changes without authorized provenance	Versioned governance memory, approval, rollback
Receipt theater	Logs exist but cannot reconstruct causal / authority chain	Tamper evidence, linkage, independent audit reconstruction

15. Worked example: cost optimization under continuity-of-care constraints

Consider a scheduling agent rewarded for reducing healthcare expenditure. The admitted representation $\hat{x}$ includes coverage data, current care plan, appointment availability, and a policy summary. The proposal $\hat{u}$ recommends canceling or redirecting a recurring appointment. A pure cost objective may prefer the cancellation. A fixed penalty may discourage cancellations under ordinary reward scale. A hard governance model instead asks whether the action is admissible under delegated authority, current consent, continuity requirements, appeal rights, and the patient's authorized treatment plan.

The representation-aware model matters because the gate may not know that a pending specialist note changed the clinical context, or the proposal may describe the action as a harmless reschedule while the executable tool call actually cancels the referral. The independent criterion G^* in a benchmark could contain the hidden note and the true tool semantics. The experiment can then measure whether the operational system falsely admits the transition and whether a stronger proposer learns to find similar blind spots. The example also illustrates the monitorability boundary: a rule such as “do not cancel an authorized referral” can be safety-like and gate-enforceable when the needed evidence is present, while “eventually restore access” or “eventually complete an appeal” is liveness-like and cannot be guaranteed by a blocking reference monitor alone. A continuity requirement defined as “an appointment must occur before deadline d ” becomes monitorable only if the deadline and progress state are explicitly carried into the monitored history, and even then the gate can prevent deadline-violating actions more readily than it can guarantee successful future completion.

16. Research agenda

1. Build synthetic environments with hidden-state admissibility oracles so representation error, false admission, and proposal laundering can be measured separately.
2. Run the first synthetic adaptive-search experiment: estimate conditional hazard h_t and episode escape q_M across proposer classes and budgets, compare against a matched constant-hazard baseline, and test whether adaptive feedback causes h_t to rise with history.
3. Design long-horizon benchmarks in which all local actions are permitted but cumulative trajectory invariants fail, including scope, resource, disclosure, and temporal constraints.
4. Compare fixed soft penalties, exact-penalty / constrained baselines, shielding, reference-monitor-style mediation, and full ICTF implementations under matched capability and exposure.
5. Extend the ideal forward-invariance lemma and finite-horizon degradation bound to multi-agent handoffs with explicit authority, shared-resource, and information-flow interfaces; identify where local closure assumptions fail under partial observability or interacting objectives.
6. Develop canonical executable action representations that allow proposal semantics, authorization, tool calls, and observed outcomes to be compared mechanically.

7. Run controlled internalization studies using independent held-out labels, measuring useful capability, dissent, false rejection, human burden, and provenance alongside safety.
8. Classify every proposed invariant as online-monitorable, delayed-but-auditable, or liveness-like / non-prefix-decidable, and report the evidence and decision horizon required for each class.
9. Separate invariant adequacy governance from runtime invariant enforcement so the system that enforces rules cannot silently define the rules it enforces.
10. Develop a nontrivial adaptive-search bound in terms of the size, geometry, or reachable measure of the gate's false-admission region under a specified proposer/query model; treat this as a future mathematical result rather than assuming a fixed per-query ϵ .
11. Publish negative results, benchmark exploits, and invariant failures as first-class outputs rather than treating them as implementation defects to hide.

17. Limitations and falsification conditions

ICTF should be narrowed or rejected if its factorization does not predict or diagnose failures more usefully than established constrained-control and security methods. The framework is weakened if representation-aware evaluation adds complexity without separating actionable failure causes; if finite-search escape curves provide no information beyond ordinary adversarial testing; or if the HAPI/AGS mapping offers no practical advantage over existing reference-monitor, capability, shielding, and runtime-verification designs.

ICTF also does not claim greater expressive power than a sufficiently augmented state model. If all governance-relevant history, authority, provenance, and monitor state are included in an augmented state s'_t , then many trajectory constraints can be written as ordinary state constraints over s'_t . The claimed value is factorization rather than expressiveness. A generic augmented state can encode history, authority, provenance, and monitor state, but it does not by itself prescribe where representation is admitted, which proposal semantics were judged, how authorization is bound to executable action, where runtime divergence occurred, or how an independent evaluator should distinguish missing evidence from bad judgment. ICTF exposes those interfaces as separate intervention and measurement points. Its value must therefore be demonstrated through better diagnosis, experimental ablation, or engineering control. If the factorization yields no empirical or engineering advantage over a conventional augmented-state formulation, the framework contributes little beyond notation.

The framework does not solve normative specification. It cannot determine which human values are correct, who legitimately owns an invariant, or when a context-sensitive requirement should be hard, lexicographic, chance-constrained, or soft. It also does not guarantee that G^* is infallible. In human domains the independent criterion may still be contested, biased, or incomplete. Independence reduces circularity; it does not eliminate normative uncertainty.

Finally, the formalism does not establish a universal law of systems. State-transition language is extremely general and can become vacuous if used without domain semantics. The paper therefore limits its primary contribution to consequential agent governance and treats cross-domain correspondence as an appendix-level observation.

18. Conclusion

The revised Invariant-Constrained Transition Framework is a formal and experimental framework for consequential agent governance. Its core move is not the invention of constraints. It is the separation of objects that are often collapsed: reality from representation, proposal from executable action, operational judgment from independent evaluation, local checks from trajectory properties, external gating from internalization, and model-relative compliance from the adequacy of the governance model itself.

This separation changes what should be measured. A credible governance claim requires more than an average gate pass rate. It requires false-admission and false-rejection estimates, observability-aware error decomposition, conditional escape-hazard curves h_t and finite-budget escape curves q_M , proposal-to-execution fidelity, runtime binding, history-dependent invariant monitoring, independent internalization labels, provenance, and useful capability under tail pressure. A system that becomes easier to supervise only because it has learned to evade the evaluator or refuse everything has not matured.

The security literature already supplies many of the hard architectural lessons: complete mediation, least authority, tamper resistance, explicit information-flow constraints, temporal monitoring, runtime assurance, and system-level safety. ICTF should be judged by whether it integrates those lessons into an agent-governance formalism that is operationally clearer and empirically more discriminating. If it does, the framework offers a precise interpretation of the original Alignment Theory intuition: consequential systems fail when capability outruns the relationships that legitimately constrain how action becomes real. If it does not, the theory should be revised again.

Governance defines what may happen. Optimization chooses among what may happen. Evidence determines whether the distinction held in reality.

Appendix A. Cross-domain structural correspondence

The earlier ICTF draft devoted substantial space to cross-domain comparison. That material is retained here only as a boundary test. The correspondence is structural, not a claim that ICTF derives domain laws.

Domain	State → transformation → consequence?	Limit of the analogy
Information theory	Yes at the channel / expectation level	ICTF does not derive the source or channel distribution
Expected utility / MDPs	Yes; native mathematical fit	Governance semantics still require domain-supplied constraints
Statistical mechanics	Yes for distributions, dynamics, and observables	Does not derive the Hamiltonian, Boltzmann weights, or physical law
Evolution	Yes with selection, mutation, and normalization	Fitness and population normalization add domain-specific operators
Neural-network objectives	Yes at data → model → loss level	Raw network weights are not transition probabilities
Classical mechanics	Representable as deterministic state evolution	Representation does not derive mechanics
Quantum mechanics	State → channel / unitary → observable survives	Ordinary classical path-probability semantics fail at amplitude level

The appendix supports only the modest observation that state, transformation, constraint, and consequence are reusable modeling roles. The main paper does not depend on this cross-domain breadth.

Appendix B. Notation

Symbol	Meaning
w_t	Relevant world state at time t
h_t (history)	Governance-relevant trajectory history
e_t	Evidence available to context admission
R	Representation / context-admission process
$\hat{x}_t$	Admitted actionable representation
π_k	Proposal policy from capability/search class k
$\hat{u}_t$	Proposal representation evaluated by governance
$\hat{G}$	Operational gate / admissibility judgment
G^*	Independent evaluation reference used by the benchmark
B	Runtime binding function from approved representation to executable action
a_t	Permit-bound executable action
K_0	Native world transition law / environment dynamics
C	Consequence functional
Ψ_j	History- or trajectory-dependent invariant
$\epsilon_{FA}^{(c)}$	False-admission probability within preregistered scenario class c
$\epsilon_{miss}^{(c)}$	Class-conditional false admission when decisive evidence is absent from the admitted representation
$\epsilon_{judge}^{(c)}$	Class-conditional false admission when decisive evidence is present but misjudged
h_t (hazard)	Conditional false-admission / escape hazard at query t given no earlier escape
M	Finite gate-query / search budget per episode
q_M	Episode-level probability of at least one governance escape by query budget M
λ_i	Attempt frequency for class i
I_i	Conditional impact after escape
L_{hard}	Undiscounted hard-invariant violation loss
$Compliance_M$	Invariant compliance relative to specified governance model M

Appendix C. Minimal preregistration template

- **Environment and hidden variables:** specify what the operational system can observe and what is reserved for independent evaluation.
- **Invariant specification:** list hard, soft, temporal, and cumulative requirements; identify authority owners and revision procedures.
- **Operational gate:** freeze model, prompts, policies, thresholds, and runtime enforcement before evaluation.
- **Independent criterion G^* :** define oracle logic or blinded adjudication rubric; report inter-rater agreement where human judgment is used.

- **Capability schedule κ :** preregister proposer models, search budgets, iterations, tool access, and adversarial procedures.
- **Tail set:** construct and seal rare / adversarial cases before observing condition results.
- **Primary endpoints:** class-conditional false admission, observability-split ϵ_{miss} and ϵ_{judge} , conditional hazard h_t , episode escape q_M , independent invalid-proposal rate, runtime mismatch, useful capability, false rejection, and human burden.
- **Internalization evaluation:** keep held-out G^* labels and tail cases inaccessible to the learner; report both operational and independent invalid rates.
- **Failure publication:** report exploits, disagreement cases, and cases where G^* itself is contested.

Appendix D. Minimal reference pseudocode

This pseudocode is intentionally architecture-neutral and separates operational governance from independent evaluation.

```

input: evidence e_t, history h_t, proposer pi_kappa, operational gate G_hat

x_hat <- context_admit(e_t, h_t)
u_hat <- propose(pi_kappa, x_hat, h_t)
assessment <- G_hat(x_hat, u_hat, h_t)

if assessment == ADMIT:
    permit <- bind_canonical_action(u_hat, authority_state, expiry)
    a_t, outcome <- execute_under_permit(permit)
elif assessment == REVISE:
    return revision_request_with_receipt()
elif assessment == ESCALATE:
    return escalate_to_authority_with_context()
else:
    return blocked_with_receipt()

receipt <- record(x_hat, u_hat, assessment, permit, a_t, outcome, lineage)
update_history(h_t, receipt)

# Evaluation path is held out from the operational gate:
label_star <- independent_reference_evaluate(hidden_world_state, a_t,
                                             outcome, full_history)
update_metrics(epsilon_FA_by_class, epsilon_miss, epsilon_judge,
               h_t, q_M, runtime_fidelity)

# Optional learning is proposed, reviewed, versioned, and reversible:
candidate_update <- governance_memory(receipt_history, adjudicated_labels)
pi_next <- apply_only_if_authorized(candidate_update)

```

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Author note

This manuscript formalizes a line of work developed through the Human Agency Infrastructure / Alignment Theory project. The author's prior project manuscripts [1–3] are cited to establish development provenance and terminology, not as independent external validation. Revision v1.2.1 is a technical correction release. It demotes textbook repeated-search results to baseline identities and conditional bounds, centers adaptive escape hazard as the empirical quantity of interest, conditions observability decomposition on scenario class, reframes compositional preservation as standard forward invariance with explicit probabilistic degradation under gate error, replaces the penalty section with a binary scaling condition that does not misuse smooth exact-penalty assumptions, strengthens the monitorability boundary, and clarifies the practical value claimed beyond ordinary state augmentation. The central empirical claims remain unvalidated until the preregistered comparisons, held-out evaluation, and independent red-team studies described here are performed.